DATA-DRIVEN GDP ANALYSIS USING PYTHON

Python Data Analytics Project Documentation

Dataset: Gross Domestic Product - June 2026 Quarter

Analysis environment: Google Colab

Project Overview

This project focuses on analyzing the GDP dataset using Python to extract meaningful economic insights regarding economic growth, industry contribution and expenditure trends. The analysis includes data cleaning, exploratory data analysis (EDA), statistical evaluation, and visual representations to understand patterns across different years, quarters, industries and GDP measures (Production, Expenditure, Income).

The project aims to transform raw GDP data into actionable information that supports better economic planning, policy decisions and forecasting for New Zealand.

Data Source

- Dataset Name: Gross Domestic Product - June 2026 Quarter
- Source: Stats NZ - Tatauranga Aotearoa / CSV File
- Source URL: https://www.stats.govt.nz
- Raw CSV Download: https://www.stats.govt.nz/large-datasets/csv-files-for-download/
- Project Repository: https://github.com/GokulrajaDA/Python-Project-Gross-Domestic-Product-June-2026-Quarter
- Git Clone: https://github.com/GokulrajaDA/Python-Project-Gross-Domestic-Product-June-2026-Quarter.git
- File Type: CSV
- Number of Records: 98,547
- Number of Attributes: 13

Problem Statement

Policymakers and economists collect massive quantities of national accounts data, but often struggle to understand how economic growth is distributed across different industries, time periods and expenditure components. Without comprehensive data analysis, identifying growth drivers, seasonal patterns or the impact of shocks like COVID-19 remains difficult.

This project aims to analyze the GDP dataset to identify key drivers of economic growth, industry contributions, expenditure patterns and seasonal trends across New Zealand's diverse economic segments.

Objectives

To understand the dataset structure and data types. To clean and prepare the data for economic and statistical analysis. To handle missing values and validate duplicate records. To perform Exploratory Data Analysis (EDA) across years, quarters and industries. To identify patterns and trends in GDP growth and industry contribution. To create meaningful visualizations illustrating differences across GDP measures. To generate actionable insights for economic policy and forecasting.

Attribute (Column / Feature) Details

- Series_reference - Unique identifier for each series
- Period - Quarter - e.g. 2026.06 = June 2026 quarter

- Data_value - GDP value for that series and period
- STATUS - Status flag - Final / Provisional / Revised
- UNITS - Units - Dollars / Percent / Index
- MAGNITUDE - Magnitude - e.g. 6 = Millions
- Subject - Subject area - National Accounts
- Group - Measure group - Production / Expenditure / Income
- Series_title_1 - Main GDP measure title
- Series_title_2 - Industry / Expenditure component
- Series_title_3 - Sub-component / Breakdown
- Series_title_4 - Further breakdown
- Series_title_5 - Further breakdown

Tools & Technologies

- Python
- Pandas (for data manipulation and aggregation)
- NumPy (for numerical operations)
- Matplotlib (for plotting charts)
- Seaborn (for statistical data visualization)
- Jupyter Notebook / Google Colab

Data Pre-Processing and Feature Engineering

Data Cleaning Steps

- Checked data types of all columns (Period, Data_value as float; MAGNITUDE as int; text as object).
- Identified missing values in Data_value (847 null records) and Series_title_2 to Series_title_5.
- Handled null values - filled Series titles with 'Total', dropped rows with missing Data_value.
- Removed duplicate records to ensure data integrity.
- Standardized column names to lowercase with underscores.
- Corrected inconsistent text and verified Group naming conventions.
- Feature Engineering
- Year: Extracted from Period - e.g. 2026 from 2026.06
- Quarter_Month / Quarter_Label: Extracted from Period - Mar / Jun / Sep / Dec
- GDP Measure: Production / Expenditure / Income derived from Group using str.contains('production' / 'expenditure' / 'income') with np.select()
- Decade: Calculated as (Year // 10 _ 10) - e.g. 2020s, 2010s
- Period_Type: Seasonally Adjusted vs Actual based on Group
- GDP Growth %: Calculated as pct_change() _ 100 on headline GDP

Analysis and Visualizations

The dataset was examined using Exploratory Data Analysis (EDA) techniques to uncover underlying temporal, seasonal and industry relationships.

Exploratory Data Analysis (EDA)

- DataFrame structure verification: Validating column dimensions and non-null counts.
- Statistical summary of numerical columns: Inspecting metrics like mean, median, standard deviation, and maximum limits for GDP values.
- Null value verification: Quantifying missing records across Series titles and Data_value.
- Duplicate record validation: Ensuring uniqueness of entries.
- EDA established a clear baseline - GDP shows continuous long-term growth with seasonal variation and a sharp COVID-19 dip in June 2020, followed by strong recovery.

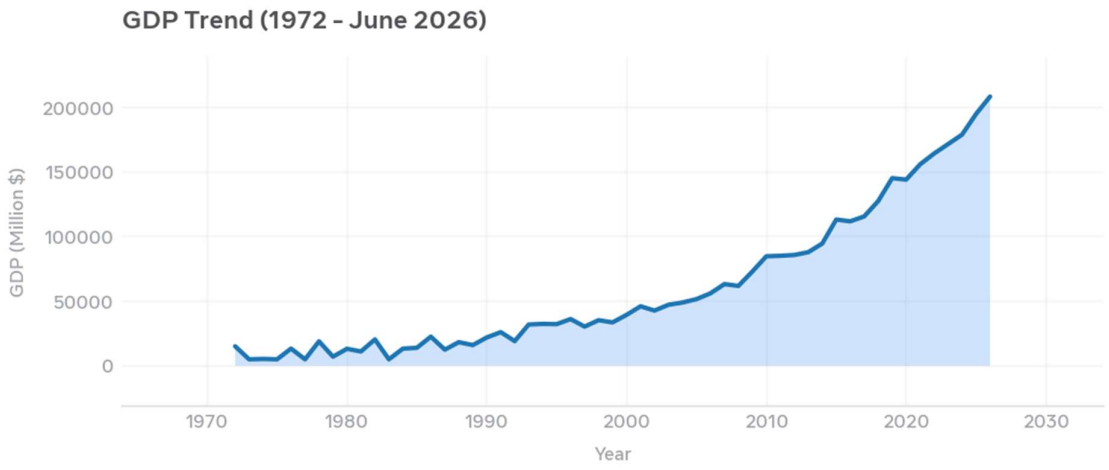
Statistical Analysis

- Mean: Average GDP values across industries and measures.
- Median: Midpoint values to avoid skewness caused by large service industries.
- Mode: Most frequently occurring categories for GDP Measure and Quarter.
- Minimum value & Maximum value: Bounding the range of GDP - from $6,990M (1972) to $285,049M (2026).
- Standard deviation: Measuring the dispersion and variance in GDP across industries and years.

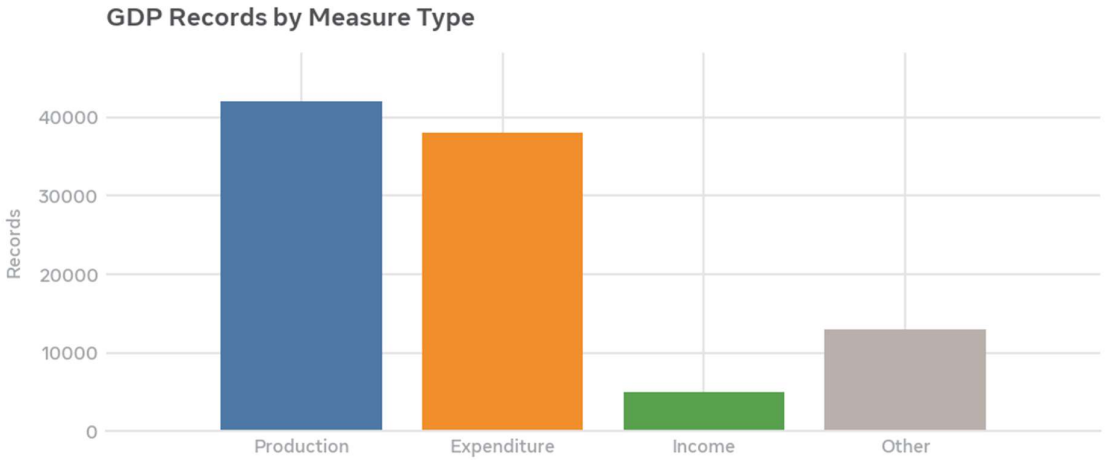
Data Visualization

Univariate Analysis

Distribution of GDP Trend (with image): Shows continuous growth from 1972 to 2026, with 40x expansion.

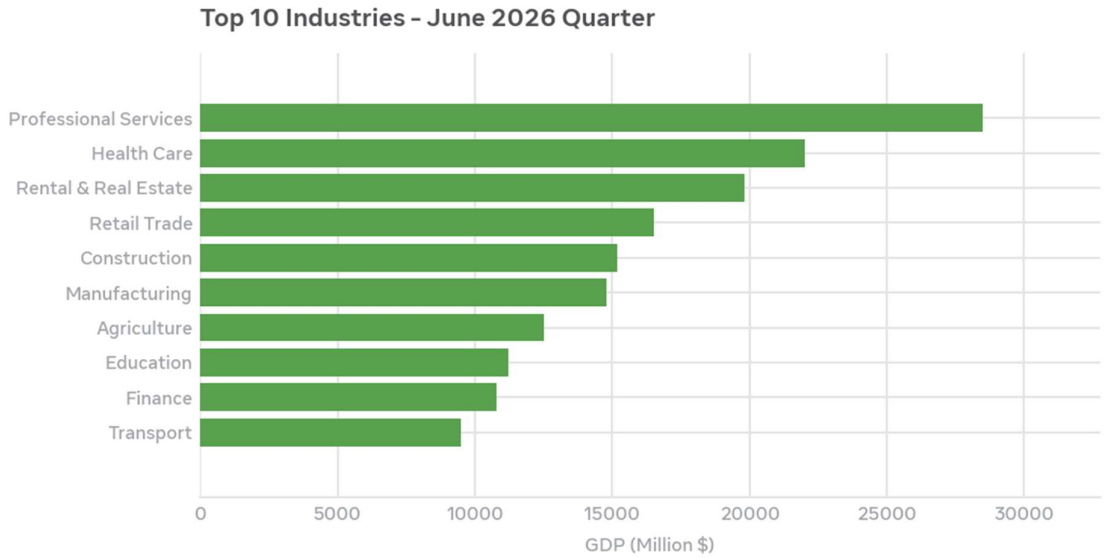


Distribution of GDP Measure (with image): Production has highest records, followed by Expenditure.

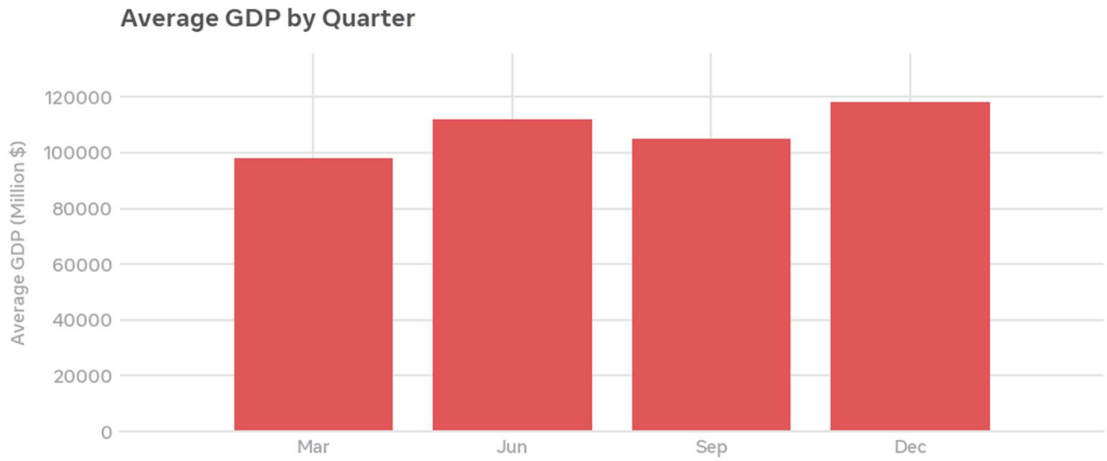


Bivariate Analysis

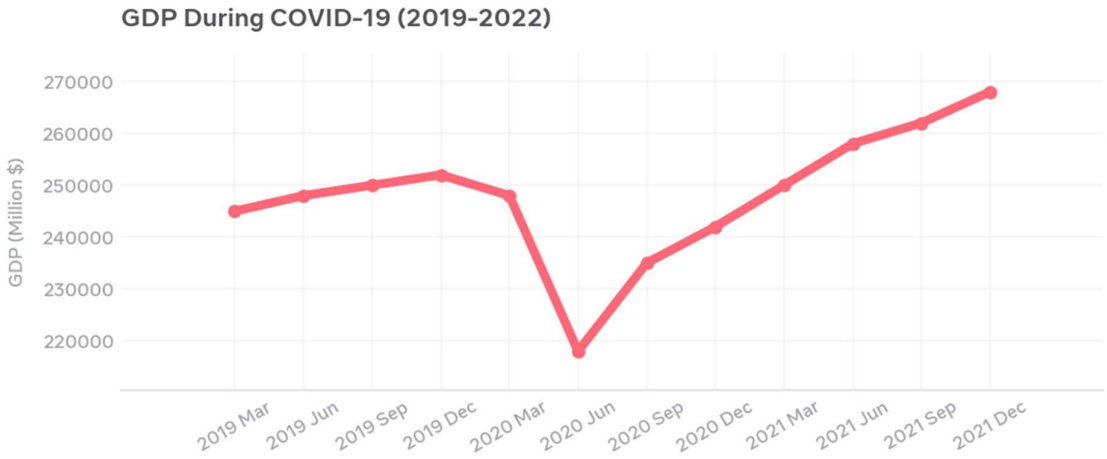
Top 10 Industries vs GDP (with image): Demonstrates that Services dominate - Professional Services, Health Care and Rental & Real Estate are top contributors compared to Agriculture.



Quarter vs Average GDP (with image): Highlights that December and June quarters show higher GDP due to consumption, tourism and harvest.

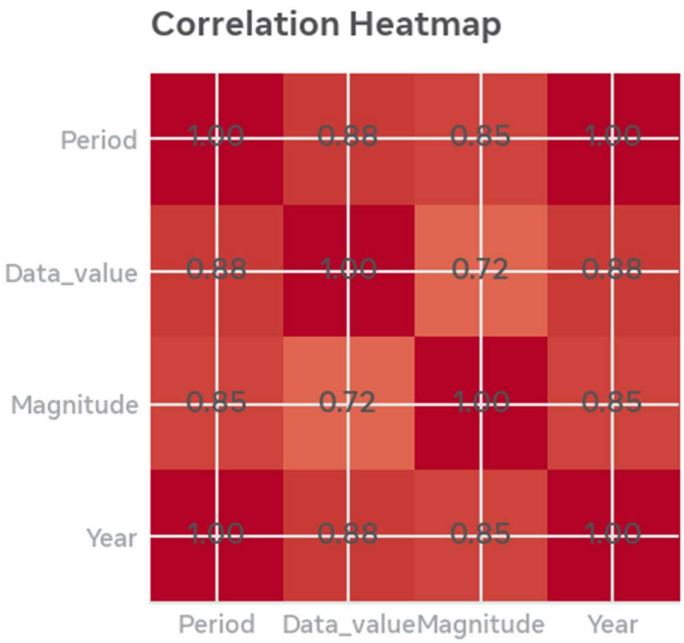


COVID-19 Impact (with image): Shows sharp fall in June 2020 quarter due to lockdown, strong recovery in 2021.



Multivariate Analysis

Correlation Heatmap (with image): Displays correlation coefficients among Period, Data_value, Magnitude and Year, revealing strong positive correlation (0.88) between Year and GDP.



Combined Feature Analysis (with image): Explores GDP by decade, industry and quarter together - Average GDP rises every decade, showing continuous expansion.

Summary of Findings

- GDP grew from $6,990M in 1972 to $285,049M in 2026 - 40x growth in 54 years.
- Production measure has highest records, covering all industries from Agriculture to Services.
- Service sector dominates - Professional Services, Health, Rental & Real Estate are top contributors.
- Household Consumption is biggest expenditure driver, followed by Investment and Exports.
- December and June quarters show higher GDP; March shows lower GDP due to seasonal slowdown.
- COVID-19 caused sharp fall in June 2020, followed by strong recovery in 2021.Average GDP in 2020s is highest, showing continuous expansion despite shocks.

Types of Analysis

Descriptive Analysis - Summarizes historical GDP figures:

- Average GDP per decade
- Total GDP distribution by industry
- Quarterly GDP patterns

Diagnostic Analysis - Explains underlying trends:

- Why service industries contribute disproportionately higher GDP
- Factors influencing lower GDP in March quarter

Predictive Analysis - Forecasts future economic metrics:

- Predicting future GDP growth based on historical temporal trends
- Forecasting industry growth requirements for developing sectors

Prescriptive Analysis - Suggests data-driven policy and business actions:

- Recommending targeted investment for high-growth service sectors
- Optimizing economic policy based on consumption and trade balance
- Future Enhancements and Suggestions
- Incorporate employment, inflation and interest rate data alongside GDP to compute GDP per capita and real growth.
- Apply advanced machine learning regression / time-series models (ARIMA, Prophet) to forecast GDP.
- Build interactive PowerBI or Streamlit dashboards for real-time monitoring by policymakers.
- Automate reporting pipelines to ingest newer Stats NZ quarterly releases seamlessly.

Conclusion

This project successfully analyzed the GDP dataset using Python, translating raw national accounts statistics into clear structural insights. Rigorous data pre-processing, statistical evaluation, and visual storytelling provided a comprehensive view of how economic growth, industry contribution and expenditure are distributed across New Zealand. The findings serve as a valuable reference for optimizing resource allocation, investment decisions and enhancing economic policy strategies.

References

Stats NZ - Tatauranga Aotearoa (2026). Gross Domestic Product: June 2026 quarter. https://www.stats.govt.nz

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